NFL-Modeling—J

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Modeling

In this document we will develop a statistical model to predict whether the offense will execute a run or a pass play. The model will include the following variables

-game_id: Unique identifier for the game

-play_id: Unique identifier for a specific play

-possession_team: Identifies the offensive team

-week: Identifies which week of the season the game was in.

-run_pass: Indicates if the play was a run of a pass (response variable)

-down: Identifies which down the offensive team was on for a particular play.

-yards_to_go: The number of yards to the opponents endzone.

-redzone: Indicates whether the offense is within 20 yards of the opponents endzone.

-sec_in_half: The number of seconds remaining on the play clock in the current half.

-offense_formation: Categorical variable indicating what formation the offense uses.

-receiver_alingment: Enumerated as 0x0, 1x0, 1x1, 2x0, 2x1, 2x2, 3x0, 3x1, 3x2 (text)

-epa_lag: Serial correllation of expected points added.

-curr_win_percentage: Win percentage for the current offensive team.

-form_change: Uses shifts to determine if the offense changed formation before the ball was snapped.

-wr_motion: Indicates if a wide receiver moved after the line was set.

-te_motion: Indicates if a tight end moved after the line was set.

-rb_motion: Indicates if a running back moved after the line was set.

-wr_atsnap: Indicates if a wide receiver was in motion at the time of the snap.

-te_atsnap: Indicates if a tight end was in motion at the time of the snap.

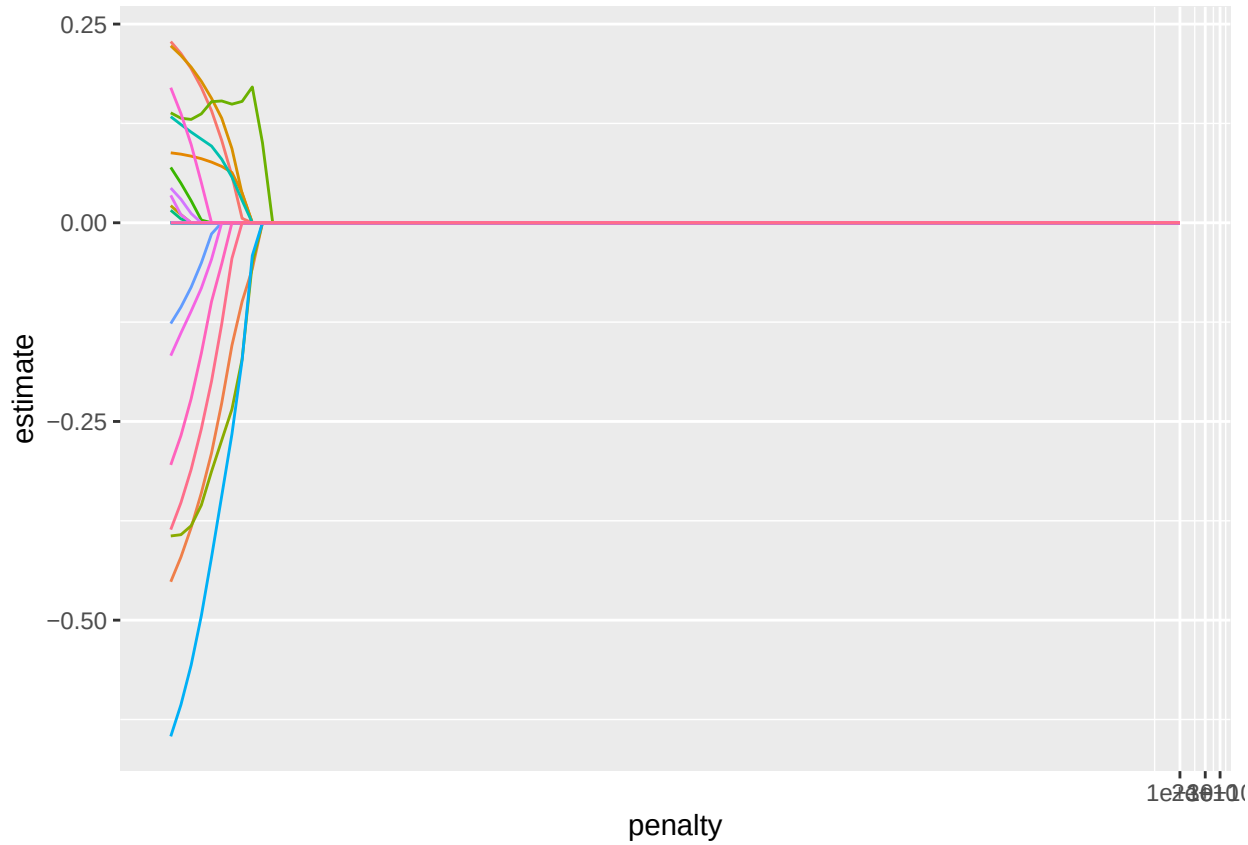
-rb_atsnap: Indicates if a running back was in motion at the time of the snap.

Model 1: Logistic (Control)

Use LASSO to perform varaible selection and then fit a logistic regression model as a kind of control to compare our other models against.

```r
#Load in the data set
nfl_df <- read.csv("data/modeling_data_frame.csv")

#Make the run pass variable a factor
nfl_df$run_pass <- as.factor(nfl_df$run_pass)

#Convert the boolean columns into numeric ones
bool_cols <- c("redzone", "wr_motion", "te_motion", "rb_motion",
               "wr_atsnap", "te_atsnap", "rb_atsnap")
nfl_df[bool_cols] <- lapply(nfl_df[bool_cols], as.numeric)

#Remove NA values
nfl_df <- nfl_df %>%
  drop_na()

#Use a lasso model to do varaible selection.

#Create a sequence of lambda values to try
lambda <- lambda <- 10^seq(-2, 10, length.out = 100)

#Create a recipe for the lasso model and normalize the predictors.
lasso_rec <- recipe(run_pass~., data=nfl_df) %>%

  #Remove unnecessary variables
  step_rm(game_id, play_id, possession_team,week) %>%

  step_novel(all_nominal_predictors()) %>%

  #Fix error for missing values by assigning them with unknown
  step_unknown(all_nominal_predictors()) %>%

  #Fix error where some variables are categorical and not numeric
  step_dummy(all_nominal_predictors()) %>%

  #Fix error from some of them having zero variance
  step_zv(all_predictors())%>%

  #Normalize the numeric predictors to have mean=0 and sd=1
  step_normalize(all_numeric_predictors())

#Create an empty data frame to store our lasso estimates
lasso_ests <- data.frame()

#Fit a lasso model for each lambda in our grid
for(lam in lambda){
  lasso_spec <- logistic_reg(mixture=1, penalty=lam)%>%
    set_mode("classification") %>%
    set_engine("glmnet")

  #Fit the model
  workflow() %>%
    add_model(lasso_spec) %>%
    add_recipe(lasso_rec) %>%
```

```
    fit(nfl_df) %>%
    tidy() %>%
    bind_rows(lasso_ests)-> lasso_ests
}

  #Create a line plot of
lasso_ests |>
  filter(term != "(Intercept)") |>
  ggplot() +
  geom_line(aes(penalty, estimate, group = term, colour = term)) +
  coord_trans(x = "log10")+
  #The legend is so large it obstructs the graph. You can comment out this line
  # and view the graph in a new window to get a better idea.
  theme(legend.position="none")
```




```
#Create the training and testing sets of data
  #Use the first 6 weeks of data for training
  #Use the last 3 weeks of dat for testing

#Grab the data
nfl_lasso_df <- nfl_df

#Make sure that run pass is a factor not a character type
nfl_lasso_df$run_pass <- as.factor(nfl_lasso_df$run_pass)
```

```r
#Encode the boolean columns as numeric
nfl_lasso_df[bool_cols] <- lapply(nfl_lasso_df[bool_cols], as.numeric)

#Create the training set
nfl_train <- nfl_lasso_df %>%
  filter(week==c(1,2,3,4,5,6))
```

```
## Warning: There was 1 warning in `filter()`.
## i In argument: `week == c(1, 2, 3, 4, 5, 6)`.
## Caused by warning in `week == c(1, 2, 3, 4, 5, 6)`:
## ! longer object length is not a multiple of shorter object length
```

```r
#Create the test set
nfl_test <- nfl_lasso_df %>%
  filter(week==c(7,8,9))
```

```
## Warning: There was 1 warning in `filter()`.
## i In argument: `week == c(7, 8, 9)`.
## Caused by warning in `week == c(7, 8, 9)`:
## ! longer object length is not a multiple of shorter object length
```

```r
#Fit a lasso model and determine the optimum lambda for varaible selection

#Create a data frame for tuning lambda
tune_df <- data.frame(lambda=lambda)

#Perform 10-fold cross-validation to get an estimate of test MSE for each
#labmda in our grid
nfl_10fld_cv <- vfold_cv(nfl_df, v=10)

#Specify the model
lasso_spec <- logistic_reg(mixture = 1, penalty = tune("lambda")) %>%
  set_mode("classification") %>%
  set_engine("glmnet")

#Create the workflow object for lasso
lasso_workflow <-workflow() %>%
  add_model(lasso_spec) %>%
  add_recipe(lasso_rec)

#Perform the cross validation
tune_results <- tune_grid( lasso_workflow,
                           resamples=nfl_10fld_cv,
                           #For some reason it works if I let the labmda values
                           #be created automatically but not when I try to use
                           #the labmda values in the tuning
                           grid=50)

#Determine the best lambda value
best_lambda <- select_best(tune_results, metric="accuracy")

#Create the lasso model with the optimum lambda
```

```r
final_lasso_workflow <- finalize_workflow(lasso_workflow,  best_lambda)

final_lasso_model <- fit(final_lasso_workflow, data=nfl_train) -> final_lasso_fit

final_model <- extract_fit_parsnip(final_lasso_model)

#Convert the coefficients to a tidy data frame
selected_variables <- tidy(final_model)

#Generate predictions on the test dataset
lasso_predictions <- predict(final_lasso_model, nfl_test, type = "class") %>%
  bind_cols(nfl_test)

#Create a confusion matrix
confusion_mat <- conf_mat(
  data = lasso_predictions,
  truth = run_pass,
  estimate = .pred_class
)

#Print the confusion matrix
print(confusion_mat)
```

```
##           Truth
## Prediction pass run
##       pass  738 189
##       run   202 486
```

```r
#Fit the logistic model on the entire data set using the varaibles from the
#lasso model.

#Look at the output from the optimal lasso model to do variable selection
selected_vars <-c("down", "yards_to_go", "yards_to_endzone", "sec_in_half",
                  "epa_lag", "curr_win_percentage", "form_change", "wr_motion",
                  "te_motion","rb_motion","wr_atsnap", "te_atsnap",
                  "offense_formation", "receiver_alignment")

#Drop the few data points that are in the testing set but not the training set
nfl_train <- nfl_train %>% filter(offense_formation %in% nfl_test$offense_formation,
                                    receiver_alignment %in% nfl_test$receiver_alignment)

nfl_test <- nfl_test %>% filter(offense_formation %in% nfl_train$offense_formation,
                                  receiver_alignment %in% nfl_train$receiver_alignment)

#Create a formula for logistic regression using selected variables
logistic_formula <- as.formula(paste("run_pass ~",
                                     paste(selected_vars, collapse = " + ")))

#Fit the logistic regression model using the selected variables
final_logistic_model <- glm(
  formula = logistic_formula,
  data = nfl_train,
  family = binomial()
```

```r
)

# Generate predictions on the test set
logistic_predictions_prob <- predict(final_logistic_model,
                                     newdata = nfl_test, type = "response")

#Convert probabilities to class labels based on a threshold of 0.5
logistic_predictions_class <- ifelse(logistic_predictions_prob > 0.5, "run", "pass")

#Add predictions to the test dataset
nfl_test <- nfl_test %>%
  mutate(predicted_run_pass = logistic_predictions_class)

#Create a confusion matrix
confusion_matrix <- table(
  Truth = nfl_test$run_pass,
  Prediction = nfl_test$predicted_run_pass
)

#Make sure that the precited pass column is also a factor like the truth col
nfl_test$predicted_run_pass <- as.factor(nfl_test$predicted_run_pass)

#Create a confusion matrix with the accuracy
confusionMatrix(nfl_test$predicted_run_pass, nfl_test$run_pass)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass  736 184
##       run   203 491
##
##                Accuracy : 0.7602
##                  95% CI : (0.7386, 0.7809)
##     No Information Rate : 0.5818
##     P-Value [Acc > NIR] : <2e-16
##
##                   Kappa : 0.5092
##
##  Mcnemar's Test P-Value : 0.3602
##
##             Sensitivity : 0.7838
##             Specificity : 0.7274
##          Pos Pred Value : 0.8000
##          Neg Pred Value : 0.7075
##              Prevalence : 0.5818
##          Detection Rate : 0.4560
##    Detection Prevalence : 0.5700
##       Balanced Accuracy : 0.7556
##
##        'Positive' Class : pass
##
```

```r
#Fit the optimal logistic regression model for each team


#Create a list to store the models for each team as well as the confusion
#matrices
model_info <- list("nfl_team" = unique(nfl_df$possession_team),
                   "train_fit" = as.list(seq(32)),
                   "prediction" = as.list(seq(32)),
                   "conf_mat" = as.list(seq(32)),
                   "m_corr"= as.list(seq(32)))

#Specify the logistic formula using the variables selected from lasso
logistic_formula <- as.formula(paste("run_pass ~",
                                     paste(selected_vars, collapse = " + ")))

#Use a for loop to create a model for each team and store it in the model info
#list
for (i in c(1:32)){

  #Grab the current team
  team<-model_info$nfl_team[i]

  #Add a print statement so we can keep track while its running
  print(paste("Working on team", team))

  #Create a data frame with the data for the current team
  team_df <- nfl_df %>%
    filter(possession_team==team) %>%
    mutate(run_pass = as.factor(run_pass))

  #troublesome receiver alignment value for denver remove it
  if(team=="DEN"){
    team_df <- team_df %>%
      filter(receiver_alignment !="2x0")
  }

  #Sort the data by week
  weeks <- sort(unique(team_df$week))

  #Split the data into training and testing
  team_train <- team_df %>%
    filter(week==c(1,2,3,4,5,6))

  team_test <- team_df %>%
    filter(week==c(7,8,9))

  #Drop the few data points that are in the testing set but not the training set
  team_train <- team_train %>%
    filter(offense_formation %in% team_test$offense_formation,
           receiver_alignment %in% team_test$receiver_alignment)

  team_test <- team_test %>%
    filter(offense_formation %in% team_train$offense_formation,
```

```r
          receiver_alignment %in% team_train$receiver_alignment)

  #Fit the logistic regression model using the selected variables
  final_logistic_model <- glm(
    formula = logistic_formula,
    data = nfl_train,
    family = binomial()
    )

  #Extract the model
  train_fit <- final_logistic_model$model

  #Add the training fit to the list
  model_info$train_fit[[i]] <- train_fit

  # Generate predictions on the test set
  logistic_predictions_prob <- predict(final_logistic_model,
                                       newdata = team_test, type = "response")

  #Convert probabilities to class labels based on a threshold of 0.5
  logistic_predictions_class <- ifelse(logistic_predictions_prob > 0.5,
                                       "run", "pass")

  #Add the predicted classes to the model info list
  model_info$predictions[[i]]<- logistic_predictions_class

  #Add predictions to the test dataset
  team_test <- team_test %>%
    mutate(predicted_run_pass = logistic_predictions_class)

  #Make sure that the precited pass column is also a factor like the truth col
  team_test$predicted_run_pass <- as.factor(team_test$predicted_run_pass)

  #Add the confusion matrix to the model info list
  model_info$conf_mat[[i]]<-confusionMatrix(team_test$predicted_run_pass,
                                            team_test$run_pass)

  #Calculate Matthews correlation coefficient (mltools)
  model_info$m_corr[[i]]<- mcc(preds=team_test$predicted_run_pass,
                               actuals= team_test$run_pass)

  #Add print statements to view the model progress
  print(team)
  print(confusionMatrix(team_test$predicted_run_pass,team_test$run_pass))
  print(model_info$m_corr[[i]])
}
```

```
## [1] "Working on team BUF"
```

```
## [1] "BUF"
## Confusion Matrix and Statistics
##
##           Reference
```

```
## Prediction pass run
##       pass   12   4
##       run     6   7
##
##                Accuracy : 0.6552
##                  95% CI : (0.4567, 0.8206)
##     No Information Rate : 0.6207
##     P-Value [Acc > NIR] : 0.4303
##
##                   Kappa : 0.2927
##
##  Mcnemar's Test P-Value : 0.7518
##
##             Sensitivity : 0.6667
##             Specificity : 0.6364
##          Pos Pred Value : 0.7500
##          Neg Pred Value : 0.5385
##              Prevalence : 0.6207
##          Detection Rate : 0.4138
##    Detection Prevalence : 0.5517
##       Balanced Accuracy : 0.6515
##
##        'Positive' Class : pass
##
## [1] 0.2956562
## [1] "Working on team LA"

## [1] "LA"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   11   5
##       run    12   7
##
##                Accuracy : 0.5143
##                  95% CI : (0.3399, 0.6862)
##     No Information Rate : 0.6571
##     P-Value [Acc > NIR] : 0.9727
##
##                   Kappa : 0.0541
##
##  Mcnemar's Test P-Value : 0.1456
##
##             Sensitivity : 0.4783
##             Specificity : 0.5833
##          Pos Pred Value : 0.6875
##          Neg Pred Value : 0.3684
##              Prevalence : 0.6571
##          Detection Rate : 0.3143
##    Detection Prevalence : 0.4571
##       Balanced Accuracy : 0.5308
##
##        'Positive' Class : pass
```

```
##
## [1] 0.05868912
## [1] "Working on team ATL"
## [1] "ATL"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   19   6
##       run     6  24
##
##                Accuracy : 0.7818
##                  95% CI : (0.6499, 0.8819)
##     No Information Rate : 0.5455
##     P-Value [Acc > NIR] : 0.0002381
##
##                   Kappa : 0.56
##
##  Mcnemar's Test P-Value : 1.0000000
##
##             Sensitivity : 0.7600
##             Specificity : 0.8000
##          Pos Pred Value : 0.7600
##          Neg Pred Value : 0.8000
##              Prevalence : 0.4545
##          Detection Rate : 0.3455
##    Detection Prevalence : 0.4545
##       Balanced Accuracy : 0.7800
##
##        'Positive' Class : pass
##
## [1] 0.56
## [1] "Working on team NO"

## [1] "NO"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   33   3
##       run     6  16
##
##                Accuracy : 0.8448
##                  95% CI : (0.7258, 0.9265)
##     No Information Rate : 0.6724
##     P-Value [Acc > NIR] : 0.002537
##
##                   Kappa : 0.6615
##
##  Mcnemar's Test P-Value : 0.504985
##
##             Sensitivity : 0.8462
##             Specificity : 0.8421
##          Pos Pred Value : 0.9167
```

```
##          Neg Pred Value : 0.7273
##              Prevalence : 0.6724
##          Detection Rate : 0.5690
##    Detection Prevalence : 0.6207
##       Balanced Accuracy : 0.8441
##
##        'Positive' Class : pass
##
## [1] 0.6657305
## [1] "Working on team CAR"

## [1] "CAR"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   20  12
##       run     5  16
##
##                Accuracy : 0.6792
##                  95% CI : (0.5368, 0.8008)
##     No Information Rate : 0.5283
##     P-Value [Acc > NIR] : 0.0186
##
##                   Kappa : 0.3659
##
##  Mcnemar's Test P-Value : 0.1456
##
##             Sensitivity : 0.8000
##             Specificity : 0.5714
##          Pos Pred Value : 0.6250
##          Neg Pred Value : 0.7619
##              Prevalence : 0.4717
##          Detection Rate : 0.3774
##    Detection Prevalence : 0.6038
##       Balanced Accuracy : 0.6857
##
##        'Positive' Class : pass
##
## [1] 0.3790877
## [1] "Working on team CLE"

## [1] "CLE"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   16   4
##       run     5  12
##
##                Accuracy : 0.7568
##                  95% CI : (0.588, 0.8823)
##     No Information Rate : 0.5676
##     P-Value [Acc > NIR] : 0.01368
```

```
##
##                   Kappa : 0.5081
##
##  Mcnemar's Test P-Value : 1.00000
##
##             Sensitivity : 0.7619
##             Specificity : 0.7500
##          Pos Pred Value : 0.8000
##          Neg Pred Value : 0.7059
##              Prevalence : 0.5676
##          Detection Rate : 0.4324
##    Detection Prevalence : 0.5405
##       Balanced Accuracy : 0.7560
##
##        'Positive' Class : pass
##
## [1] 0.5088846
## [1] "Working on team CHI"

## [1] "CHI"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   20  16
##       run     4  21
##
##                Accuracy : 0.6721
##                  95% CI : (0.54, 0.7869)
##     No Information Rate : 0.6066
##     P-Value [Acc > NIR] : 0.17992
##
##                   Kappa : 0.3685
##
##  Mcnemar's Test P-Value : 0.01391
##
##             Sensitivity : 0.8333
##             Specificity : 0.5676
##          Pos Pred Value : 0.5556
##          Neg Pred Value : 0.8400
##              Prevalence : 0.3934
##          Detection Rate : 0.3279
##    Detection Prevalence : 0.5902
##       Balanced Accuracy : 0.7005
##
##        'Positive' Class : pass
##
## [1] 0.3982193
## [1] "Working on team SF"

## [1] "SF"
## Confusion Matrix and Statistics
##
##           Reference
```

```
## Prediction pass run
##       pass   20   4
##       run     5   9
##
##                Accuracy : 0.7632
##                  95% CI : (0.5976, 0.8856)
##     No Information Rate : 0.6579
##     P-Value [Acc > NIR] : 0.1138
##
##                   Kappa : 0.4834
##
##  Mcnemar's Test P-Value : 1.0000
##
##             Sensitivity : 0.8000
##             Specificity : 0.6923
##          Pos Pred Value : 0.8333
##          Neg Pred Value : 0.6429
##              Prevalence : 0.6579
##          Detection Rate : 0.5263
##    Detection Prevalence : 0.6316
##       Balanced Accuracy : 0.7462
##
##        'Positive' Class : pass
##
## [1] 0.484182
## [1] "Working on team PIT"

## [1] "PIT"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   25   6
##       run     4   8
##
##                Accuracy : 0.7674
##                  95% CI : (0.6137, 0.8824)
##     No Information Rate : 0.6744
##     P-Value [Acc > NIR] : 0.1259
##
##                   Kappa : 0.4501
##
##  Mcnemar's Test P-Value : 0.7518
##
##             Sensitivity : 0.8621
##             Specificity : 0.5714
##          Pos Pred Value : 0.8065
##          Neg Pred Value : 0.6667
##              Prevalence : 0.6744
##          Detection Rate : 0.5814
##    Detection Prevalence : 0.7209
##       Balanced Accuracy : 0.7167
##
##        'Positive' Class : pass
```

```
##
## [1] 0.4528748
## [1] "Working on team CIN"

## [1] "CIN"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   34   7
##       run     3  14
##
##                Accuracy : 0.8276
##                  95% CI : (0.7057, 0.9141)
##     No Information Rate : 0.6379
##     P-Value [Acc > NIR] : 0.001303
##
##                   Kappa : 0.6107
##
##  Mcnemar's Test P-Value : 0.342782
##
##             Sensitivity : 0.9189
##             Specificity : 0.6667
##          Pos Pred Value : 0.8293
##          Neg Pred Value : 0.8235
##              Prevalence : 0.6379
##          Detection Rate : 0.5862
##    Detection Prevalence : 0.7069
##       Balanced Accuracy : 0.7928
##
##        'Positive' Class : pass
##
## [1] 0.618279
## [1] "Working on team DET"

## [1] "DET"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   23   2
##       run     9  16
##
##                Accuracy : 0.78
##                  95% CI : (0.6404, 0.8847)
##     No Information Rate : 0.64
##     P-Value [Acc > NIR] : 0.02468
##
##                   Kappa : 0.56
##
##  Mcnemar's Test P-Value : 0.07044
##
##             Sensitivity : 0.7188
##             Specificity : 0.8889
```

```
##          Pos Pred Value : 0.9200
##          Neg Pred Value : 0.6400
##              Prevalence : 0.6400
##          Detection Rate : 0.4600
##    Detection Prevalence : 0.5000
##       Balanced Accuracy : 0.8038
##
##        'Positive' Class : pass
##
## [1] 0.5833333
## [1] "Working on team PHI"

## [1] "PHI"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   15   4
##       run     3  13
##
##                Accuracy : 0.8
##                  95% CI : (0.6306, 0.9156)
##     No Information Rate : 0.5143
##     P-Value [Acc > NIR] : 0.0004642
##
##                   Kappa : 0.599
##
##  Mcnemar's Test P-Value : 1.0000000
##
##             Sensitivity : 0.8333
##             Specificity : 0.7647
##          Pos Pred Value : 0.7895
##          Neg Pred Value : 0.8125
##              Prevalence : 0.5143
##          Detection Rate : 0.4286
##    Detection Prevalence : 0.5429
##       Balanced Accuracy : 0.7990
##
##        'Positive' Class : pass
##
## [1] 0.6000032
## [1] "Working on team IND"

## [1] "IND"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   37   5
##       run     6   8
##
##                Accuracy : 0.8036
##                  95% CI : (0.6757, 0.8977)
##     No Information Rate : 0.7679
```

```
##     P-Value [Acc > NIR] : 0.3257
##
##                   Kappa : 0.4634
##
##  Mcnemar's Test P-Value : 1.0000
##
##             Sensitivity : 0.8605
##             Specificity : 0.6154
##          Pos Pred Value : 0.8810
##          Neg Pred Value : 0.5714
##              Prevalence : 0.7679
##          Detection Rate : 0.6607
##    Detection Prevalence : 0.7500
##       Balanced Accuracy : 0.7379
##
##        'Positive' Class : pass
##
## [1] 0.463967
## [1] "Working on team HOU"

## [1] "HOU"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   21  11
##       run     4  13
##
##                Accuracy : 0.6939
##                  95% CI : (0.5458, 0.8175)
##     No Information Rate : 0.5102
##     P-Value [Acc > NIR] : 0.007038
##
##                   Kappa : 0.3839
##
##  Mcnemar's Test P-Value : 0.121335
##
##             Sensitivity : 0.8400
##             Specificity : 0.5417
##          Pos Pred Value : 0.6562
##          Neg Pred Value : 0.7647
##              Prevalence : 0.5102
##          Detection Rate : 0.4286
##    Detection Prevalence : 0.6531
##       Balanced Accuracy : 0.6908
##
##        'Positive' Class : pass
##
## [1] 0.4008302
## [1] "Working on team NE"

## [1] "NE"
## Confusion Matrix and Statistics
##
```

```
##           Reference
## Prediction pass run
##       pass   28   7
##       run     5  10
##
##                Accuracy : 0.76
##                  95% CI : (0.6183, 0.8694)
##     No Information Rate : 0.66
##     P-Value [Acc > NIR] : 0.08699
##
##                   Kappa : 0.4495
##
##  Mcnemar's Test P-Value : 0.77283
##
##             Sensitivity : 0.8485
##             Specificity : 0.5882
##          Pos Pred Value : 0.8000
##          Neg Pred Value : 0.6667
##              Prevalence : 0.6600
##          Detection Rate : 0.5600
##    Detection Prevalence : 0.7000
##       Balanced Accuracy : 0.7184
##
##        'Positive' Class : pass
##
## [1] 0.4514452
## [1] "Working on team MIA"

## [1] "MIA"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   21   6
##       run    12  14
##
##                Accuracy : 0.6604
##                  95% CI : (0.5173, 0.7848)
##     No Information Rate : 0.6226
##     P-Value [Acc > NIR] : 0.3391
##
##                   Kappa : 0.3176
##
##  Mcnemar's Test P-Value : 0.2386
##
##             Sensitivity : 0.6364
##             Specificity : 0.7000
##          Pos Pred Value : 0.7778
##          Neg Pred Value : 0.5385
##              Prevalence : 0.6226
##          Detection Rate : 0.3962
##    Detection Prevalence : 0.5094
##       Balanced Accuracy : 0.6682
##
```

```
##        'Positive' Class : pass
##
## [1] 0.3261463
## [1] "Working on team NYJ"

## [1] "NYJ"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   26   7
##       run     5  16
##
##                Accuracy : 0.7778
##                  95% CI : (0.644, 0.8796)
##     No Information Rate : 0.5741
##     P-Value [Acc > NIR] : 0.00143
##
##                   Kappa : 0.5404
##
##  Mcnemar's Test P-Value : 0.77283
##
##             Sensitivity : 0.8387
##             Specificity : 0.6957
##          Pos Pred Value : 0.7879
##          Neg Pred Value : 0.7619
##              Prevalence : 0.5741
##          Detection Rate : 0.4815
##    Detection Prevalence : 0.6111
##       Balanced Accuracy : 0.7672
##
##        'Positive' Class : pass
##
## [1] 0.5420179
## [1] "Working on team BAL"
## [1] "BAL"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   16   4
##       run     6  34
##
##                Accuracy : 0.8333
##                  95% CI : (0.7148, 0.9171)
##     No Information Rate : 0.6333
##     P-Value [Acc > NIR] : 0.0005889
##
##                   Kappa : 0.6341
##
##  Mcnemar's Test P-Value : 0.7518296
##
##             Sensitivity : 0.7273
##             Specificity : 0.8947
```

```
##          Pos Pred Value : 0.8000
##          Neg Pred Value : 0.8500
##              Prevalence : 0.3667
##          Detection Rate : 0.2667
##    Detection Prevalence : 0.3333
##       Balanced Accuracy : 0.8110
##
##        'Positive' Class : pass
##
## [1] 0.6358508
## [1] "Working on team NYG"
## [1] "NYG"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   15  11
##       run     4   7
##
##                Accuracy : 0.5946
##                  95% CI : (0.421, 0.7525)
##     No Information Rate : 0.5135
##     P-Value [Acc > NIR] : 0.2058
##
##                   Kappa : 0.1802
##
##  Mcnemar's Test P-Value : 0.1213
##
##             Sensitivity : 0.7895
##             Specificity : 0.3889
##          Pos Pred Value : 0.5769
##          Neg Pred Value : 0.6364
##              Prevalence : 0.5135
##          Detection Rate : 0.4054
##    Detection Prevalence : 0.7027
##       Balanced Accuracy : 0.5892
##
##        'Positive' Class : pass
##
## [1] 0.1950445
## [1] "Working on team TEN"

## [1] "TEN"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   14   5
##       run     8  18
##
##                Accuracy : 0.7111
##                  95% CI : (0.5569, 0.8363)
##     No Information Rate : 0.5111
##     P-Value [Acc > NIR] : 0.005111
```

```
##
##                   Kappa : 0.4202
##
##  Mcnemar's Test P-Value : 0.579100
##
##             Sensitivity : 0.6364
##             Specificity : 0.7826
##          Pos Pred Value : 0.7368
##          Neg Pred Value : 0.6923
##              Prevalence : 0.4889
##          Detection Rate : 0.3111
##    Detection Prevalence : 0.4222
##       Balanced Accuracy : 0.7095
##
##        'Positive' Class : pass
##
## [1] 0.4240305
## [1] "Working on team JAX"

## [1] "JAX"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   36  12
##       run     3   9
##
##                Accuracy : 0.75
##                  95% CI : (0.6214, 0.8528)
##     No Information Rate : 0.65
##     P-Value [Acc > NIR] : 0.06561
##
##                   Kappa : 0.3902
##
##  Mcnemar's Test P-Value : 0.03887
##
##             Sensitivity : 0.9231
##             Specificity : 0.4286
##          Pos Pred Value : 0.7500
##          Neg Pred Value : 0.7500
##              Prevalence : 0.6500
##          Detection Rate : 0.6000
##    Detection Prevalence : 0.8000
##       Balanced Accuracy : 0.6758
##
##        'Positive' Class : pass
##
## [1] 0.4193139
## [1] "Working on team WAS"

## [1] "WAS"
## Confusion Matrix and Statistics
##
##           Reference
```

```
## Prediction pass run
##       pass   24   9
##       run     3  14
##
##                Accuracy : 0.76
##                  95% CI : (0.6183, 0.8694)
##     No Information Rate : 0.54
##     P-Value [Acc > NIR] : 0.001124
##
##                   Kappa : 0.5074
##
##  Mcnemar's Test P-Value : 0.148915
##
##             Sensitivity : 0.8889
##             Specificity : 0.6087
##          Pos Pred Value : 0.7273
##          Neg Pred Value : 0.8235
##              Prevalence : 0.5400
##          Detection Rate : 0.4800
##    Detection Prevalence : 0.6600
##       Balanced Accuracy : 0.7488
##
##        'Positive' Class : pass
##
## [1] 0.5235176
## [1] "Working on team KC"

## [1] "KC"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   25   2
##       run    11   7
##
##                Accuracy : 0.7111
##                  95% CI : (0.5569, 0.8363)
##     No Information Rate : 0.8
##     P-Value [Acc > NIR] : 0.9479
##
##                   Kappa : 0.3434
##
##  Mcnemar's Test P-Value : 0.0265
##
##             Sensitivity : 0.6944
##             Specificity : 0.7778
##          Pos Pred Value : 0.9259
##          Neg Pred Value : 0.3889
##              Prevalence : 0.8000
##          Detection Rate : 0.5556
##    Detection Prevalence : 0.6000
##       Balanced Accuracy : 0.7361
##
##        'Positive' Class : pass
```

```
##
## [1] 0.3855678
## [1] "Working on team ARI"
## [1] "ARI"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   35  12
##       run     2   6
##
##                Accuracy : 0.7455
##                  95% CI : (0.61, 0.8533)
##     No Information Rate : 0.6727
##     P-Value [Acc > NIR] : 0.15713
##
##                   Kappa : 0.3257
##
##  Mcnemar's Test P-Value : 0.01616
##
##             Sensitivity : 0.9459
##             Specificity : 0.3333
##          Pos Pred Value : 0.7447
##          Neg Pred Value : 0.7500
##              Prevalence : 0.6727
##          Detection Rate : 0.6364
##    Detection Prevalence : 0.8545
##       Balanced Accuracy : 0.6396
##
##        'Positive' Class : pass
##
## [1] 0.3716909
## [1] "Working on team LAC"

## [1] "LAC"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   26   4
##       run     3   7
##
##                Accuracy : 0.825
##                  95% CI : (0.6722, 0.9266)
##     No Information Rate : 0.725
##     P-Value [Acc > NIR] : 0.1042
##
##                   Kappa : 0.5484
##
##  Mcnemar's Test P-Value : 1.0000
##
##             Sensitivity : 0.8966
##             Specificity : 0.6364
##          Pos Pred Value : 0.8667
```

```
##          Neg Pred Value : 0.7000
##              Prevalence : 0.7250
##          Detection Rate : 0.6500
##    Detection Prevalence : 0.7500
##       Balanced Accuracy : 0.7665
##
##        'Positive' Class : pass
##
## [1] 0.549532
## [1] "Working on team LV"

## [1] "LV"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   32   2
##       run     6  12
##
##                Accuracy : 0.8462
##                  95% CI : (0.7192, 0.9312)
##     No Information Rate : 0.7308
##     P-Value [Acc > NIR] : 0.03743
##
##                   Kappa : 0.6414
##
##  Mcnemar's Test P-Value : 0.28884
##
##             Sensitivity : 0.8421
##             Specificity : 0.8571
##          Pos Pred Value : 0.9412
##          Neg Pred Value : 0.6667
##              Prevalence : 0.7308
##          Detection Rate : 0.6154
##    Detection Prevalence : 0.6538
##       Balanced Accuracy : 0.8496
##
##        'Positive' Class : pass
##
## [1] 0.6519457
## [1] "Working on team MIN"

## [1] "MIN"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   22   3
##       run     6  10
##
##                Accuracy : 0.7805
##                  95% CI : (0.6239, 0.8944)
##     No Information Rate : 0.6829
##     P-Value [Acc > NIR] : 0.1181
```

```
##
##                   Kappa : 0.5226
##
##  Mcnemar's Test P-Value : 0.5050
##
##             Sensitivity : 0.7857
##             Specificity : 0.7692
##          Pos Pred Value : 0.8800
##          Neg Pred Value : 0.6250
##              Prevalence : 0.6829
##          Detection Rate : 0.5366
##    Detection Prevalence : 0.6098
##       Balanced Accuracy : 0.7775
##
##        'Positive' Class : pass
##
## [1] 0.5293838
## [1] "Working on team GB"
## [1] "GB"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   30  14
##       run     4   8
##
##                Accuracy : 0.6786
##                  95% CI : (0.5404, 0.7971)
##     No Information Rate : 0.6071
##     P-Value [Acc > NIR] : 0.16934
##
##                   Kappa : 0.2674
##
##  Mcnemar's Test P-Value : 0.03389
##
##             Sensitivity : 0.8824
##             Specificity : 0.3636
##          Pos Pred Value : 0.6818
##          Neg Pred Value : 0.6667
##              Prevalence : 0.6071
##          Detection Rate : 0.5357
##    Detection Prevalence : 0.7857
##       Balanced Accuracy : 0.6230
##
##        'Positive' Class : pass
##
## [1] 0.2927858
## [1] "Working on team DAL"

## [1] "DAL"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
```

```
##       pass   15   2
##       run     6   9
##
##                Accuracy : 0.75
##                  95% CI : (0.566, 0.8854)
##     No Information Rate : 0.6562
##     P-Value [Acc > NIR] : 0.1768
##
##                   Kappa : 0.49
##
##  Mcnemar's Test P-Value : 0.2888
##
##             Sensitivity : 0.7143
##             Specificity : 0.8182
##          Pos Pred Value : 0.8824
##          Neg Pred Value : 0.6000
##              Prevalence : 0.6562
##          Detection Rate : 0.4688
##    Detection Prevalence : 0.5312
##       Balanced Accuracy : 0.7662
##
##        'Positive' Class : pass
##
## [1] 0.5067912
## [1] "Working on team TB"
```

```
## [1] "TB"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   47   2
##       run     4   9
##
##                Accuracy : 0.9032
##                  95% CI : (0.8012, 0.9637)
##     No Information Rate : 0.8226
##     P-Value [Acc > NIR] : 0.06004
##
##                   Kappa : 0.6905
##
##  Mcnemar's Test P-Value : 0.68309
##
##             Sensitivity : 0.9216
##             Specificity : 0.8182
##          Pos Pred Value : 0.9592
##          Neg Pred Value : 0.6923
##              Prevalence : 0.8226
##          Detection Rate : 0.7581
##    Detection Prevalence : 0.7903
##       Balanced Accuracy : 0.8699
##
##        'Positive' Class : pass
##
```

```
## [1] 0.6942197
## [1] "Working on team SEA"

## [1] "SEA"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   19  11
##       run    12  13
##
##                Accuracy : 0.5818
##                  95% CI : (0.4411, 0.7135)
##     No Information Rate : 0.5636
##     P-Value [Acc > NIR] : 0.4483
##
##                   Kappa : 0.1538
##
##  Mcnemar's Test P-Value : 1.0000
##
##             Sensitivity : 0.6129
##             Specificity : 0.5417
##          Pos Pred Value : 0.6333
##          Neg Pred Value : 0.5200
##              Prevalence : 0.5636
##          Detection Rate : 0.3455
##    Detection Prevalence : 0.5455
##       Balanced Accuracy : 0.5773
##
##        'Positive' Class : pass
##
## [1] 0.1539504
## [1] "Working on team DEN"

## [1] "DEN"
## Confusion Matrix and Statistics
##
##           Reference
## Prediction pass run
##       pass   22   6
##       run     2  12
##
##                Accuracy : 0.8095
##                  95% CI : (0.6588, 0.914)
##     No Information Rate : 0.5714
##     P-Value [Acc > NIR] : 0.001029
##
##                   Kappa : 0.6
##
##  Mcnemar's Test P-Value : 0.288844
##
##             Sensitivity : 0.9167
##             Specificity : 0.6667
##          Pos Pred Value : 0.7857
```

```
##          Neg Pred Value : 0.8571
##              Prevalence : 0.5714
##          Detection Rate : 0.5238
##    Detection Prevalence : 0.6667
##       Balanced Accuracy : 0.7917
##
##        'Positive' Class : pass
##
## [1] 0.6123724
```

```r
#Save the data
saveRDS(model_info, "Logistic Model Info by Team.rds")
```